BHDD: A Burmese Handwritten Digit Dataset

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***Abstract*—We introduce the Burmese Handwritten Digit Dataset (BHDD), a collection of 87,561 grayscale images of handwritten Burmese digits in ten classes. Each image is 28×28 pixels, following the MNIST format. The training set has 60,000 samples split evenly across classes; the test set has 27,561 samples with class frequencies as they arose during collection. Over 150 people of different ages and backgrounds contributed samples. We analyze the dataset's class distribution, pixel statistics, and morphological variation, and identify digit pairs that are easily confused due to the round shapes of the Myanmar script. Simple baselines (an MLP, a two-layer CNN, and an improved CNN with batch normalization and augmentation) reach 99.40%, 99.75%, and 99.83% test accuracy respectively. BHDD is available under CC BY-SA 4.0 at https://github.com/baseresearch/BHDD.**

***Index Terms*—handwritten digit recognition, benchmark dataset, Myanmar script, Burmese digits, optical character recognition**

I. Introduction

MNIST [1] set the standard for handwritten digit recognition in 1998: 70,000 images of Latin digits, each 28×28 grayscale pixels, with a fixed train/test split. It has been used in thousands of studies since. EMNIST [2] extended the same format to Latin letters, and Kuzushiji-MNIST [3] showed that Japanese cursive characters produce very different error patterns than Latin digits; different scripts need their own benchmarks.

Myanmar has roughly 55 million people. Burmese, the main language, is spoken natively by over 33 million. The Myanmar script, called *sar-lone* (literally "round script"), developed its circular letterforms because it was traditionally written on palm leaves, where straight strokes would tear through the leaf. The digits carry over this roundness: they are built from curves, loops, and arcs, and several pairs look similar enough to confuse classifiers.

Until now, no public benchmark dataset existed for Burmese handwritten digits. We release BHDD to fill that gap. The dataset contains 87,561 labeled images from over 150 contributors. We also provide a detailed analysis of the data and three baseline models.

II. Related Work

MNIST [1] has 70,000 handwritten Latin digit images and remains the most common benchmark for digit recognition. Its fixed format (28×28 grayscale, integer labels, standard split) made it easy to compare results across models. EMNIST [2] added handwritten letters in the same format.

Kuzushiji-MNIST [3] took the same approach for cursive Japanese characters and found that the error patterns looked nothing like those on Latin digits—different scripts need their own benchmarks. Since then, handwritten digit datasets have appeared for Kannada [4], Bengali [5], Arabic [6], and Persian [7].

Myanmar script recognition has been studied since at least 2005, when Sandar [8] compared off-line handwriting and print recognition using hidden Markov models. Lwin and Wu [9] later used K-means clustering with a CNN to handle visual similarity among Myanmar's glyphs, and Aung et al. [10] released myOCR, an OCR pipeline for machine-printed Myanmar text. However, none of these efforts produced a publicly available benchmark for handwritten digits. BHDD is the first.

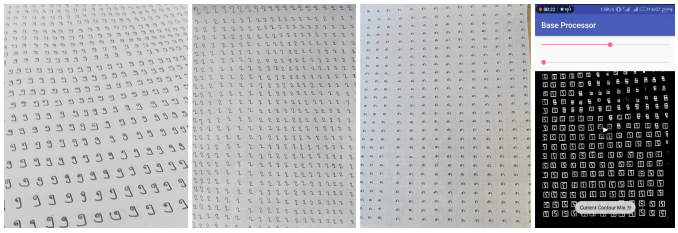

Fig. 1. Sample collection sheets on plain paper (left) and grid paper (middle two), and the mobile preprocessing app (right) that let contributors verify digit extraction before submission.

III. Dataset

A. Collection Methodology

We organized a community collection effort through the Expa.AI Research Team. Over 150 people wrote Burmese digits on plain A4 paper (both white and yellow sheets), each filling ten or more pages (one per digit class), with roughly 500–600 digits per page. In total, approximately 2,500 sheets were collected. Contributors came from Taungoo Computer University (volunteers and interns), St. Augustine / B.E.H.S (2) Kamayut (high school students), and friends and family of the research team. Ages ranged from teenagers to people in their 50s, with occupations including clerks, programmers, and students. Fig. 1 shows representative collection sheets.

The majority of sheets were photographed with phone cameras rather than scanned. Variable lighting, angles, and

This dataset was collected and analyzed in 2019.

camera quality made automated digit extraction difficult, so we developed an Android application (Fig. 1, right) that applied adaptive thresholding and contour detection in real time, capturing per-photo preprocessing parameters as metadata. Contributors could adjust settings and retake photos until the extracted digits were clean. Because extraction yield from phone photos was low, the team collected substantially more sheets than the final dataset required.

The extraction pipeline used OpenCV: convert to grayscale, binarize with adaptive thresholding (using the per-photo parameters captured by the app), find contours to locate each digit, crop to the bounding box, center in a 28×28 frame, and normalize pixel intensity.

Quality assurance was a two-stage process: the 20-member annotation team inspected samples distributed via a chatbot-based review tool, and two data engineers then performed a final verification pass over the complete dataset. Mislabeled, illegible, and duplicate images were removed at each stage. All 87,561 images in the final dataset are verified unique: no exact duplicates exist within or across the training and test sets.

Approximately 120 contributors were assigned to the training set and the remaining roughly 30 to the test set, so no writer's handwriting appears in both splits. Most contributors were based in Yangon, with smaller groups from Mandalay, Nay Pyi Taw, Shan State, and the United States.

B. Data Format and Structure

Each sample is a 28×28 grayscale image (uint8, values 0–255) with an integer label from 0 to 9. The dataset is available in Python pickle and gzip-compressed IDX formats; the IDX files work directly with any MNIST data loader. Fig. 2 shows samples from all ten classes.

C. Class Distribution

The dataset contains 87,561 images: 60,000 for training and 27,561 for testing. The split is by contributor; all pages from a given person go entirely to one set, so no writer's handwriting appears in both. The training set is downsampled to exactly 6,000 samples per class. The test set is left unbalanced (Fig. 3), with per-class counts ranging from 6,856 (class 0) to 389 (class 9), reflecting how often each digit appeared in the collected sheets.

D. Visual and Statistical Analysis

Pixel intensity. Mean pixel intensity per class ranges from 12.5 (class 2) to 26.5 (class 0). Ink coverage (the fraction of non-zero pixels) ranges from 30.4% (class 2, a thin hook) to 56.8% (class 0, a full circle). Per-class distributions are shown in Fig. 4.

Mean images. Averaging all 6,000 training samples per class gives the images in Fig. 5. Each class has a distinct shape: class 0 is a ring, class 1 is an open arc, class 8 is a spiral. The averages are sharp rather than blurry, meaning writers agree on the basic form of each digit.

Variance. Fig. 6 maps where pixel values vary most across writers. High-variance spots sit at stroke endpoints and junctions, where individual handwriting styles diverge most. Complex digits like class 8 show wide variance regions; simple ones like class 2 show a narrow band.

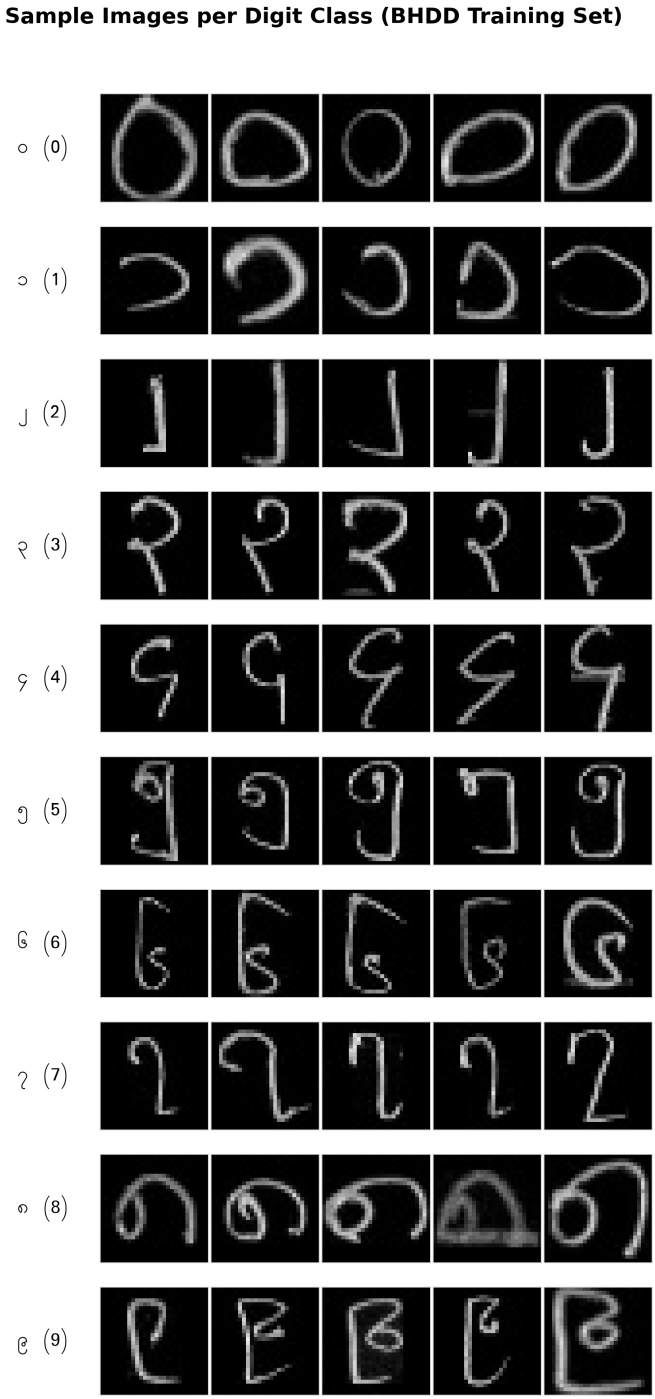

Fig. 2. Samples from all ten Burmese digit classes (rows: 0–9).

Writing styles. Fig. 7 shows individual samples from four classes with high within-class variation (0, 3, 5, 8). Stroke thickness, curvature, and how far the loop closes all differ from writer to writer.

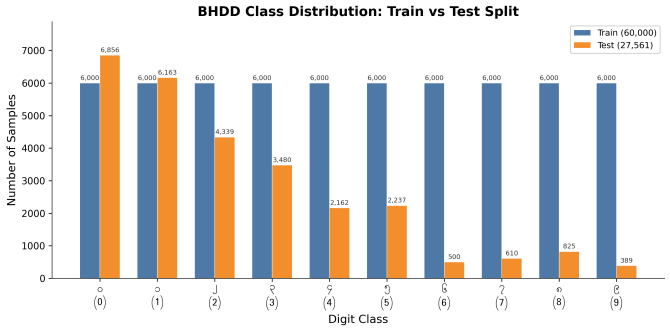

Fig. 3. Training set (balanced, 6,000 per class) vs. test set (unbalanced, 6,856 down to 389).

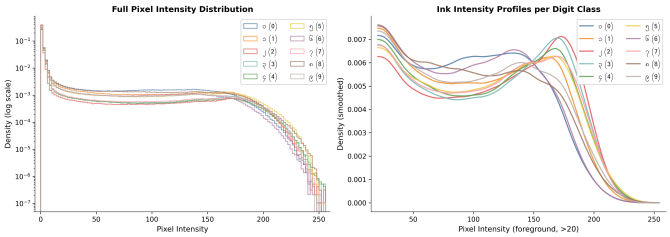

Fig. 4. Pixel intensity distributions per class. Left: log-scale histogram. Right: per-class kernel density estimates.

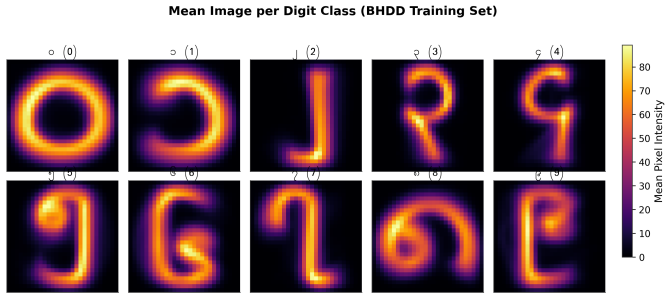

Fig. 5. Mean image per class (6,000 training samples each). Sharp shapes show consistent digit forms across writers.

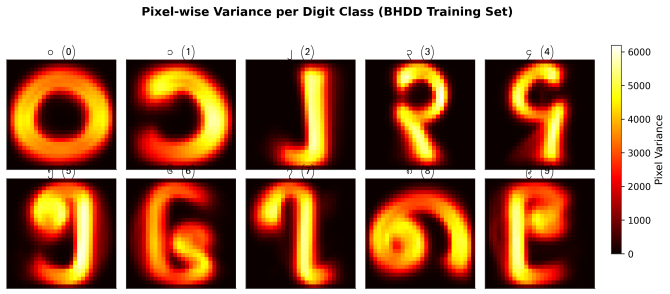

Fig. 6. Per-class pixel variance. Bright areas mark where writers differ most.

E. Script-Specific Challenges

Because Myanmar digits are built from round strokes, some pairs look very similar. We identified the worst pairs from our improved CNN's confusion matrix (Section IV-C) and show them in Fig. 8.

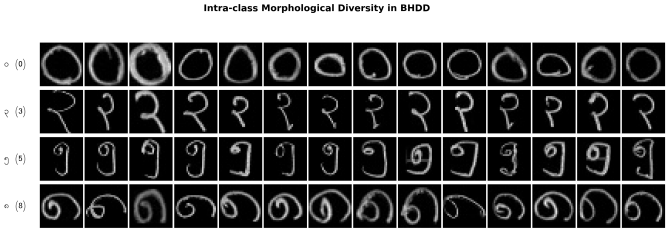

Fig. 7. Writing style variation within classes 0, 3, 5, and 8.

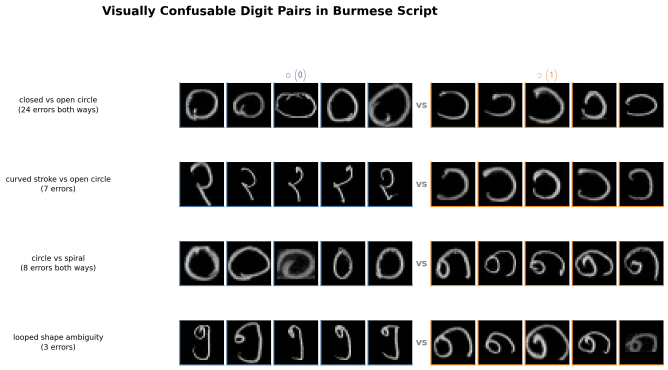

Fig. 8. Commonly confused digit pairs with error counts from the improved CNN.

Digits 0 and 1 are the hardest pair: 24 misclassifications between them (in both directions combined). The only difference is whether the circle is closed or has a small gap. Digits 0 and 8 share a round outer shape but differ inside (plain ring vs. spiral), with 8 errors combined. Digit 3 gets mistaken for 1 seven times when its tail is faint and only the curved head shows. Digits 5 and 8 sometimes overlap because both have looped forms (3 errors).

When digits share curved sub-strokes, small differences in pen pressure or whether someone fully closes a loop can shift a sample's appearance toward a neighboring class.

IV. Baseline Experiments

We trained three models to establish initial performance on the dataset. All runs used seed 42.

A. Models

MLP. Two hidden layers (256, 128 units), ReLU, Adam, trained on flattened 784-d input with early stopping (10% validation split, up to 50 epochs). Implemented in scikit-learn [11].

CNN. Two conv layers (32 and 64 filters, 3×3, ReLU, 2×2 max-pool), dropout (0.25 spatial, 0.5 dense), one FC layer (128 units), 10-way output. Trained 15 epochs with Adam ($\text{lr} = 10^{-3}$) and cross-entropy. Implemented in PyTorch [12].

Improved CNN. Three conv layers (32, 32, 64 filters) with batch normalization after each, two max-pool stages, and a 128-unit FC layer (431K parameters, barely more

TABLE I
BASELINE RESULTS ON THE BHDD TEST SET

Model	Accuracy	F1	Precision	Recall
MLP	0.9940	0.9905	0.9876	0.9934
CNN	0.9975	0.9964	0.9959	0.9970
Improved CNN	0.9983	0.9980	0.9972	0.9988

F1, Precision, and Recall are macro-averaged.

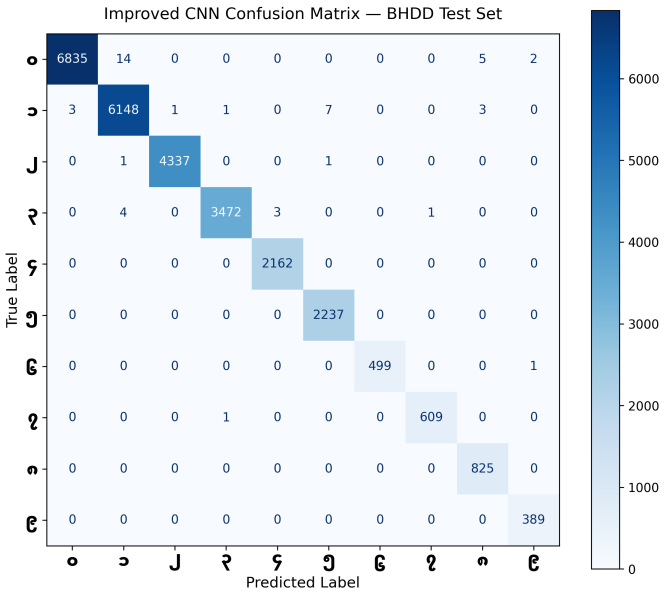

Fig. 9. Improved CNN confusion matrix. Nearly all off-diagonal entries are zero. The 0–1 pair dominates the remaining errors.

than the baseline's 421K). Trained 25 epochs with on-the-fly augmentation (rotation $\pm 15°$, translation ± 2 px, scale 0.9–1.1$\times$), cosine learning-rate annealing, and light weight decay.

B. Results

All three models exceed 99% accuracy with standard training procedures (Table I). The improved CNN reaches 99.83% with a macro F1 of 0.998, cutting misclassifications by about a third compared to the baseline CNN while adding only 10K parameters.

C. Error Analysis

The improved CNN's confusion matrix (Fig. 9) has nearly all off-diagonal entries at zero. Of 27,561 test samples, only 47 are misclassified. The errors cluster around digits 0 and 1 (a closed versus open circle), which account for about half of all mistakes, consistent with the structural similarity between these two classes.

V. AVAILABILITY

BHDD is at https://github.com/baseresearch/BHDD under CC BY-SA 4.0, in pickle and IDX formats. The repository includes exploration scripts, baseline code, and usage examples.

VI. CONCLUSION AND FUTURE WORK

We have released BHDD, a dataset of 87,561 Burmese handwritten digit images from over 150 contributors. The training set is balanced; the test set preserves the class frequencies as they arose during collection. Simple baselines reach up to 99.83% accuracy, and the remaining errors arise from visual similarity between round digits that share curved sub-strokes.

Next, we plan to expand the dataset to cover Myanmar consonants and characters, collect word- and sentence-level handwriting, and study how test-set imbalance affects evaluation.

ACKNOWLEDGMENTS

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